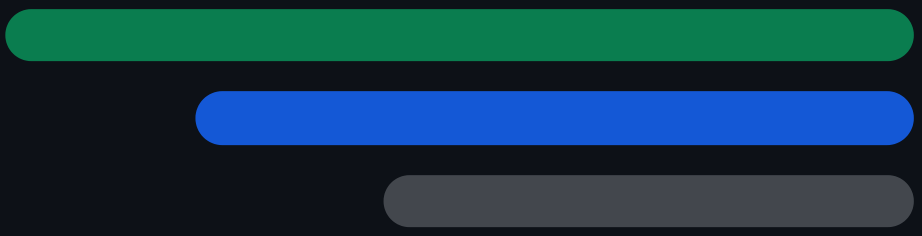
ULTIMATE APP · BORACAY

Technical Documentation

Architecture, modules, money flows, integrations, deployment and testing of the Ultimate App platform.



Document	Technical Documentation, edition 2026.10
Date	15 October 2026
Platform	PHP 8.1+ · MySQL / MariaDB (InnoDB) · Progressive Web App
Hosting	Hostinger (shared hosting, HTTPS required)
Source	ailevoler/serialnumber · `ultimateapp-agents-portal/`
Audience	Developers, system administrators, finance and operations

Contents

1	System overview	Apps, users, technology stack, size
2	Architecture	Layers, folders, request flow, settings, storage
3	Money model and ledger	Credits, BCash, centavos, double-entry, idempotency
4	Payment methods	Credits, BCash, Pay Cash switch, charges
5	URide	Vehicles incl. Boat, fares, dispatch, rentals, self-drive, fleets
6	Marketplace	UPass, UGo, ULocal, UEat
7	Cancellation fees and reasons	Rules, percentages, ledger
8	Wallet, transfers and exchange	Buy Credits, BCash, P2P/P2M, currency exchange
9	Bills, load and cash-in	SUNIWAY Partner API
10	Identity: KYC, liveness, Face Login	MediaPipe, face-api, review flow
11	Portals	Admin, Merchant Portal and MCTC, Driver App, Agents Portal, UBarangay
12	Security	Sessions, CSRF, rate limits, files, secrets
13	Deployment	Server requirements, migrations, configuration, cron
14	Testing	Test suites and fake provider servers
15	Operations and known limits	Backups, reconciliation, open items

1 System overview

Ultimate App is a mobile-first super-app for Boracay. One PHP code base and one MySQL database serve five web applications: the customer app, the URide Driver App, the Merchant Portal, the Agents Portal and the Admin Panel.

227	~20,100	58	24
PHP files (excluding tests and vendor)	lines of PHP	database tables	automated test suites

1.1 Applications and users

Application	URL	Users	Main purpose
Customer app (PWA)	`/`	Residents and tourists	Wallets (Credits, BCash), URide, UPass/UGo/ULocal/UEat, bills and load, QR payments, exchange, UBarangay, KYC, Face Login
URide Driver App (PWA)	`/driver/`	Drivers, boat operators, vehicle owners	Sign-up, go online, receive and run rides and rentals, self-drive hand-over, wallet, top-ups, payouts
Merchant Portal (PWA)	`/merchant-portal/`	Businesses	Onboarding, payment QR, listings and orders, UEat restaurant, payouts, MCTC top-up center
Agents Portal	`/agent-portal/`	Referral agents	Referral link/QR, Master and Sub-Agent teams, commissions, payouts
Admin Panel	`/admin/`	Super Admin, Finance, Support, Barangay Officer	CRM, ERP (payments, finance, ledger, settlements), URide, marketplace, KYC, settings, audit log

1.2 Technology stack

Area	Technology
Server language	PHP 8.1 or newer (tested on 8.3), `declare(strict_types=1)` in libraries; no framework
Database	MySQL / MariaDB, InnoDB, `utf8mb4`; PDO with exceptions and prepared statements
Front end	Server-rendered HTML, plain CSS and vanilla JavaScript; installable PWA (service worker, manifests)
Maps	Google Maps JavaScript API, Places API (New), Geocoding API (optional)
Face and liveness	MediaPipe Face Landmarker (Apache-2.0) and face-api (MIT), both self-hosted under `assets/vendor/`
QR codes	qrcode-generator 1.4.4 (MIT), html5-qrcode 2.3.8 (Apache-2.0)
Payments in	PayMongo QR Ph (Payment Intents and Hosted Checkout, signed webhooks)
Bills, load, e-wallet cash-in	SUNIWAY Partner API

Area	Technology
Exchange rates	Frankfurter (ECB) with currency-api fallback, or a manual rate
Time zone	Asia/Manila in PHP and `+08:00` in every MySQL session

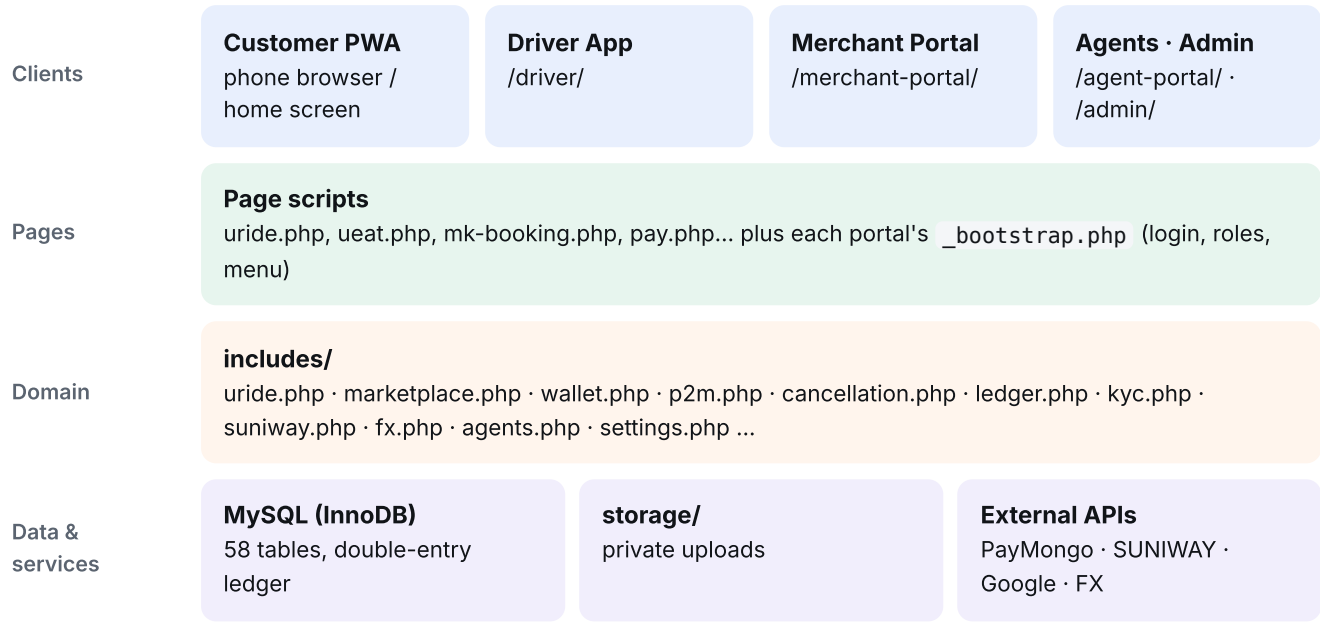
1.3 Customer services

Service	What it does	Paid with
URide	Point-to-point rides (Motorcycle, E-Trike, Car, Van, **Boat**), Rent a Ride with driver, Self-drive	Credits, BCash, Cash*
UPass	Airport transfers, island/day passes, event passes, entrance fees	Credits, BCash, on the day*
UGo	Tours and activities with time slots	Credits, BCash, on the day*
ULocal	Local services (spa, photography, laundry, guides…)	Credits, BCash, on the day*
UEat	Food delivery and pick-up from merchant restaurants	Credits, BCash
UBills / ULoad / UCash In	Bills payment, prepaid load, send to GCash/Maya and other e-wallets	Credits
Pay / Scan QR	Pay merchants (P2M) and people (P2P)	Credits, BCash
Exchange	Buy and sell USD and about 20 other currencies	BCash
UBarangay	Barangay permits and certificates with QR verification (switchable)	Credits, BCash, QR Ph

* Cash and pay-on-the-day are available only while Admin keeps **Pay Cash** on (chapter 4).

2 Architecture

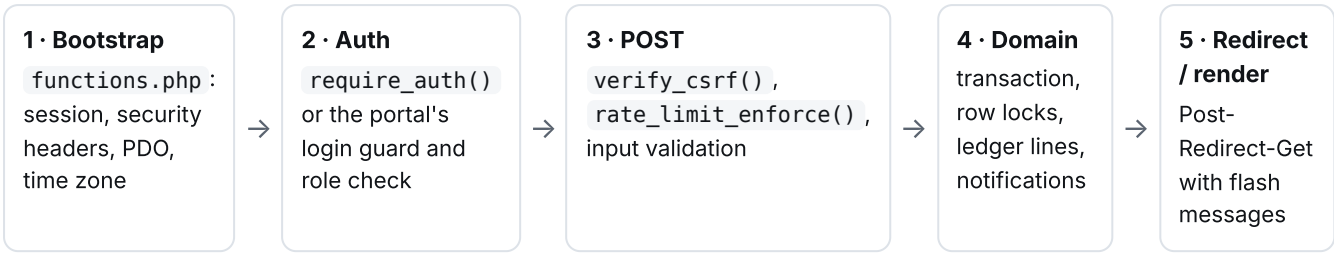
A classic page-per-file PHP application. Each page includes a shared bootstrap, does its own POST handling (CSRF-checked), calls domain functions in `includes/`, and renders HTML. All business rules live in the domain libraries so that pages, cron jobs and tests use the same code.



2.1 Folder layout

Path	Contents
`/*.php`	Customer pages (dashboard, uride, ueat, upass/ugo/ulocal, pay, bcash-*, kyc, face-login, ubills...)
`includes/`	Domain libraries and shared layout (`functions.php` bootstrap, `header.php`, `footer.php`, `menu.php`)
`admin/`	Admin Panel; `_bootstrap.php` defines roles, permissions and menu
`driver/`	URide Driver App (own login, manifest, JSON `api.php` for live feed and GPS)
`merchant-portal/`	Merchant Portal (onboarding, listings, orders, payouts, MCTC top-up)
`agent-portal/`	Agents Portal (referrals, team, earnings, payouts)
`assets/`	CSS, JavaScript, images, PWA icons, `vendor/mediapipe` (~22 MB), `vendor/face-api` (~14 MB)
`config/`	`config.php` (DB), `paymongo.php`, `google_maps.php`, `app_secret.php` (generated encryption key), `storage.php` (optional)
`database/`	`schema.sql` and 37 migration files, plus `ALL_UPDATES_2026-09.sql`
`storage/`	Private uploads: `kyc/`, `licenses/`, `driver_docs/`, `market/`, `brgy/`, `cache/` (blocked from the web)
`cron/`	`fx-refresh.php` (hourly exchange-rate refresh)
`tests/`	CLI test suites and fake provider servers (never deployed)

2.2 Request flow



Domain functions throw `InvalidArgumentException` for user-facing errors (shown as alerts) and other exceptions for system faults (logged, generic message shown). Every money change runs inside one InnoDB transaction.

2.3 Settings

- Stored in `app_settings` (`setting_key`, `setting_value`, `is_secret`). Read with `setting()` / `setting_int()`; written with `setting_save()`, which records the admin and audit trail.
- `includes/settings.php` holds the schema: type, limits and label for every key (e.g. `uride.boat.per_km_centavos`, `cancel.ueat_pct`).
- Secret values (PayMongo and SUNIWAY keys) are encrypted with the key in `config/app_secret.php`. Back up that file with the site, or the secrets must be entered again.
- Most settings are edited in **Admin › Settings** (Super Admin only): Payments, Charges & limits, Google Maps, URide.

2.4 File storage

Uploads are saved on disk, not in the database, with random names, and served only through access-checked PHP scripts (e.g. `market-photo.php`, `admin/kyc-file.php`). Images are re-encoded on upload, which strips EXIF/GPS data. To keep files safe across redeploys, copy `config/storage.php.example` to `config/storage.php` and point it to a folder outside `public_html`. **Admin › System › Files & storage** runs a write test and lists records whose files are missing.

3 Money model and ledger

Two customer wallets, one rule: every money movement is posted to a balanced double-entry ledger in the same database transaction as the balance change.

Credits

The main in-app balance. **1 Credit = ₱1.00.** Stored in `users.credits` (decimal pesos). Bought with QR Ph or at an MCTC top-up center; spent on every service. Ledger account `user_credits:ID`.

BCash

A separate peso wallet in `boracay_cash_wallets.balance_centavos`. Converted 1:1 from/to Credits with no fee; used for transfers, payments and currency exchange. Ledger account `user_cash:ID`.

3.1 Amounts

- All new money columns are **integer centavos** (`*_centavos`). No floating-point arithmetic is used for money.
- Percentages are computed with integer basis points and rounded once (e.g. commission `intdiv(fare × bp + 5000, 10000)`).
- URide fares and rental prices are rounded **up** to the whole peso.

3.2 The ledger

`ledger_post($pdo, $group, $event, $lines)` inserts lines into `ledger_entries`. Each line is `[account, debit, credit]`. The function refuses any entry whose debits and credits differ, or that is zero. The `group` ties the lines to a business reference (e.g. a ride code, `RF-…` for a refund, `CF-…` for a cancellation fee).

Account	Meaning
`user_credits:ID`	Customer Credits
`user_cash:ID`	Customer BCash
`merchant_payable:ID`	Owed to a merchant (paid out in Settlements)
`driver_wallet:ID`	URide driver wallet
`uride_escrow`	URide fares on hold until the trip ends
`uride_deposit`	Self-drive deposits held
`platform_sales:<service>`	Ultimate App sales (its own UPass/UGo/ULocal listings)
`fee_revenue:<kind>`	Charge and commission revenue
`agent_payable:ID · agent_commission_expense`	Agent commissions owed and their expense
`paymongo_clearing · bank_clearing`	Money in transit with PayMongo or the bank
`settlement_payable:…`	Payouts requested but not yet transferred
`mctc_merchant_due:ID · mctc_collections:ID · office_cash`	Cash collected for top-ups
`suniway_payable`	Owed to SUNIWAY for bills, load and cash-in

Account	Meaning
`fx_usd_book · fx_book:<ccy> · fx_fee_revenue`	Customer foreign-currency holdings (PHP value) and exchange fees
`brgy_payable:<barangay>`	Permit fees owed to each barangay
`adjustments`	Manual admin adjustments

Admin › Finance shows liabilities, revenue, a trial balance and reconciliation checks (for example, every driver wallet and every merchant MCTC due must equal its ledger balance). **Admin › General ledger** lists and exports entries.

3.3 Safety against double charging

- **Request keys.** Booking and payment forms carry a random one-time key; replays return the original result instead of charging again.
- **Row locks.** Balances, rides, orders and top-ups are read with `SELECT … FOR UPDATE` inside the transaction (e.g. first driver to accept wins).
- **Unique constraints.** Payment IDs, references, receipt numbers per merchant and approval keys are unique in the database.
- **Webhooks** are verified by signature and matched to the stored order (amount, currency, mode, session) before settling.

4 Payment methods

Admin controls which payment methods customers see. Credits are always on; BCash and Pay Cash have their own switches.

Admin › Settings › Charges & limits › Payment methods

Method	Switch	Covers
Credits	Always on	Every service
BCash	`pay_bcash_enabled()`	URide (rides, rentals, self-drive), UPass, UGo, ULocal, UEat, QR payments
Pay Cash	`pay_cash_enabled()`	Cash to the URide driver; "pay on the day" for UPass, UGo and ULocal

- When Pay Cash is off, every payment is held in the app and refunded automatically on cancel, decline or expiry.
- URide has its own Credits/Cash/BCash switches; a method shows only when both the app-wide and the URide switch are on.
- Existing bookings keep the method they were made with.

4.1 Implementation

Function (includes/wallet.php)	Purpose
`wallet_debit_tx($pdo,$uid,$source,$amount,$title,$evenWhenOff=false)`	Debit Credits or BCash inside the caller's transaction; refuses insufficient balance. `$evenWhenOff` lets a cancellation fee be taken even if BCash payments were switched off later.
`wallet_credit_tx(…)`	Credit (refund) the same wallet
`wallet_account() · wallet_balance() · wallet_label()`	Ledger account name, balance and display label for a source
`p2m_pay_tx() · p2m_refund_tx() (includes/p2m.php)`	Pay a merchant from Credits or BCash with the configured charge, and refund it in full

4.2 Merchant charges

Paying a merchant (QR or marketplace) applies the charge in **Charges & limits**: a percentage plus a fixed amount, separately for Credits and BCash, and who pays it (customer = added on top; merchant = deducted, like MDR). Each payment stores the charge it was made with. Listings run by Ultimate App have no charge and post to `platform_sales`.

5 URide

Ride-hailing, rentals with a driver, and self-drive rentals, with a driver wallet, live dispatch and full ledger posting. Library: `includes/uride.php` and `includes/uride_selfdrive.php`.

5.1 Vehicle types Boat added

Defined in `uride_vehicles()`. The flag `ride` allows point-to-point booking; `rent` allows hourly rental (with driver or self-drive). `uride_rentable_vehicles()` returns only rentable types.

Key	Label	Seats	Ride	Rent	Self-drive license
`e_bike`	E-Bike	1 rider	—	Yes (self-drive by default)	Verified ID only (Admin can require A/A1)
`motorcycle`	Motorcycle	1 passenger	Yes	Yes	Code A / A1
`e_trike`	E-Trike	3 passengers	Yes	Yes	Code A / A1
`car`	Car	4 passengers	Yes	Yes	Code B
`van`	Van	10–12 passengers	Yes	Yes	Code B / B1
`boat`	**Boat**	Up to 10	Yes	**No — point to point only**	— (never self-drive)

> **Boat rules**
> Boat appears only in Book a Ride. Rental and self-drive quotes are always disabled, `uride_book_rental()` refuses it ("Boat is for point-to-point rides only"), fleet units ignore it, and Admin's rental and self-drive tables skip it. Only drivers registered with vehicle type Boat receive Boat requests. Operators enter their MARINA / boat captain license and the boat registration number.

5.2 Fares (Book a Ride)

Upfront and fixed at booking: **fare = max(minimum, base + per-km × (road km − km in base) + night surcharge)**, rounded up to the peso. Road km = straight-line distance × road factor (default 1.3). Set in **Admin › Settings › URide › Fare matrix**. Motorcycle and E-Trike use the October 2026 pricing (₱50 for the first 2 km, then ₱25 per km, so 4 km = ₱100; `uride_pricing_2026_10_migration.sql`); the other rows are code defaults:

Vehicle	Base (km incl.)	Per km after	Minimum
Motorcycle	₱50 (2 km)	₱25	₱50
E-Trike	₱50 (2 km)	₱25	₱50
Car	₱100 (1 km)	₱25	₱100
Van	₱300 (1 km)	₱40	₱300
Boat	₱500 (1 km)	₱100	₱500

5.3 Ride lifecycle and dispatch

Requested fare held (Credits/BCash) in `uride_escrow` → **Broadcast** online drivers of that type within the radius (3 km), GPS ≤ 2 min old → **Accepted** first accept wins (row lock) → **Arrived → In progress** live position for the passenger → **Completed** driver wallet + fare − commission

- Unaccepted requests expire after the timeout (default 5 min; rentals 10 min) and are refunded.
- Drivers need ₱50 in the wallet to go online, and the minimum top-up is ₱50; cash rides deduct the trip fee from the wallet.
- Trip fee: a fixed **₱10 per completed ride** (default), or a percentage of the fare; never more than the fare. **Rent a Ride** (with driver and self-drive) instead pays a **15% commission** of each completed rental and of late-return fees (`uride.rental_commission_bp`).
- Passengers rate the driver 1–5 stars. Drivers can skip a request or give an accepted ride back before pick-up.

5.4 Rent a Ride (with driver)

Packages of 1–12 hours or a whole day (default 8 h), now or scheduled up to 7 days ahead. Price = per hour × hours (at least the minimum), never more than the whole-day price. Scheduled rentals appear to drivers 60 minutes before the start.

Vehicle	Per hour	Min hours	Whole day
E-Bike (off by default)	₱150	1	₱800
Motorcycle	₱300	1	₱1,800
E-Trike	₱500	1	₱3,000
Car	₱800	2	₱5,000
Van	₱1,500	3	₱9,000

5.5 Self-drive and fleets

- The customer needs a verified ID (switchable), a driver's license on file that covers the vehicle and does not expire before the end, and must accept the self-drive rules. Paid with Credits or BCash; price plus refundable deposit are held.
- Owners enter how many units of each type they have (`uride_fleet`). Customers see "N available"; "Now" bookings are refused when nothing is free.
- Hand-over: the owner confirms the license and ID check; the rental clock starts. Return: good condition (deposit back), late (hourly rate per started hour after the grace period, default 15 min), or damage claim decided by Admin.

Vehicle	Per hour	Min	Whole day	Deposit
E-Bike	₱100	1 h	₱600	₱1,000
Motorcycle	₱200	2 h	₱1,200	₱3,000
E-Trike	₱350	2 h	₱2,000	₱2,000

Vehicle	Per hour	Min	Whole day	Deposit
Car	₱600	4 h	₱3,500	₱5,000
Van	₱1,200	4 h	₱7,000	₱10,000

5.6 Places and maps

Pick-up and drop-off suggestions combine the customer's recent places, Admin-managed landmarks (**Admin › URide › URide places**), approved merchants near Boracay and Google Places. GPS "precise location" keeps the best fix for up to 20 s; pins are draggable and stored with 7 decimals.

5.7 Driver wallet

Event	Driver wallet	Ledger
Trip reward at a milestone	+ bonus	driver_reward_expense → driver_wallet
Ride completed (Credits/BCash)	+ fare − commission	uride_escrow → driver_wallet + fee_revenue:uride
Ride completed (cash)	− commission	driver_wallet → fee_revenue:uride
Passenger cancellation fee	+ fee	uride_escrow → user wallet (refund) + driver_wallet (fee)
Top-up: QR Ph / MCTC / BCash	+ amount	paymongo_clearing / mctc_merchant_due / user_cash → driver_wallet
Payout requested → paid	− amount	driver_wallet → settlement_payable → bank_clearing

6 Marketplace: UPass, UGo, ULocal, UEat

Merchants sell bookings and food through the Merchant Portal; Ultimate App can also run its own listings. Library: `includes/marketplace.php`, page helpers in `includes/market_page.php` and `includes/market_forms.php`.

6.1 Bookings (UPass, UGo, ULocal)

- Customer chooses date, start time (time slots with a guest capacity), guests or quantity, and for ULocal the address.
- Statuses: **pending → confirmed → completed**, or **declined / cancelled**. Listings can confirm instantly.
- Paid in the app (Credits or BCash) or on the day (when Pay Cash is on). Merchant bookings use `p2m_pay_tx()`; Ultimate App listings post to `platform_sales:<service>`.
- Each listing sets a free-cancellation cut-off (`cancel_hours` before the start).

6.2 UEat orders



- Restaurant profile: cuisine, cover, prep time, delivery fee, minimum order, hours (Philippine time), delivery and/or pick-up, open/close switch.
- Delivery is done by the restaurant itself; there is no rider dispatch yet.
- Merchant order board refreshes automatically; every status change creates a customer notification.

6.3 Moderation

Admin › UGo · ULocal · UEat: review new and edited listings and restaurants, manage any listing, feature listings, and confirm, complete or cancel bookings and orders (always with a full refund). Listing photos are stored in `storage/market/` and served by `market-photo.php`.

6.4 Agent commissions

Customers referred by an agent earn the agent a commission on completed URide trips, paid UEat orders and completed UPass/UGo/ULocal bookings (chapter 11.4).

7 Cancellation fees and reasons New

Customers must give a reason every time they cancel. A configurable percentage of what they paid in the app is kept as a fee when they cancel late. Library: `includes/cancellation.php`.

7.1 Settings

Admin › Settings › Charges & limits › Cancellation fees (0–100%, up to 2 decimals; 0% = free)

Setting key	Default	Applies when	Fee goes to
`cancel.uride_pct`	20%	After the free minutes once a driver accepted, or after the driver arrived	Driver wallet
`cancel.uride_free_min`	2 min	Free window after the driver accepts	—
`cancel.booking_pct`	25%	After the listing's cut-off and before the start	Merchant (or Ultimate App sales)
`cancel.ueat_pct`	30%	While the order is accepted or preparing	Restaurant

7.2 When customers can cancel

Service	Free	With fee	Not allowed
URide	Searching; first free minutes after accept	After the free minutes; driver arrived	Trip started
UPass / UGo / ULocal	Pending; confirmed before the cut-off	Between the cut-off and the start	After the start
UEat	Before the restaurant accepts	Accepted or preparing	Ready or on the way

- Only in-app payments (Credits or BCash) carry a fee. Cash rides and pay-on-the-day bookings never do.
- Declines by a driver, merchant or restaurant, and cancellations by Admin, are always refunded in full.
- Reasons come from `cancel_reasons($area)`; picking **Other** requires typed text. Reason and fee are stored (`cancel_reason`, `cancel_fee_centavos`) and shown to the customer, the driver or merchant (Orders › History) and Admin.

7.3 Ledger

- **URide:** `uride_release_hold($pdo,$ride,'cancelled',$fee)` moves held − fee back to the customer's wallet and the fee to `driver_wallet:ID` (group `RF-…`); the driver's wallet history shows "Cancellation fee".
- **Merchant bookings and UEat:** full refund first, then `mk_cancel_fee_to_merchant_tx()` moves the fee from the customer's wallet to `merchant_payable:ID` (group `CF-…`).
- **Ultimate App listings:** only paid − fee is refunded; the fee stays in `platform_sales`.

8 Wallet, transfers and exchange

8.1 Buy Credits

- **QR Ph (PayMongo):** in-app QR via Payment Intent (`topup-qr.php`), with hosted checkout as fallback. Credited only by a signed webhook (`payment.paid` or `checkout_session.payment.paid`) or by an authenticated status check that matches session, reference, mode and exact amount. QR expires after 30 minutes. A ₱5.00 service fee per top-up is configurable.
- **MCTC (cash):** the customer shows a request QR at an MCTC top-up center (an approved merchant). The merchant scans it, enters a receipt number (unique per merchant) and confirms; Credits are added immediately and the cash becomes the merchant's MCTC due, deducted from the next payout.

8.2 BCash

- Convert Credits ⇄ BCash 1:1, no fee, atomic with a unique request key.
- **Send / Receive / My QR:** BCash QR codes (with the BCash logo) always send BCash; Credits QR codes always pay Credits. ₱1–₱10,000 per transfer, max 20 sends per hour, no fee.
- When KYC is required for customers, unverified accounts cannot send Credits or BCash or use UCash In.

8.3 P2M and P2P payments

Scan QR detects merchant QR codes (`UM-…` , P2M) and person QR codes (`UA-…` , P2P). P2M shows the charge and total before paying, with Credits or BCash.

8.4 Currency exchange

- USD/PHP is the main pair; about 20 more currencies can be enabled in **Admin › ERP › Currency exchange**. Each currency has its own wallet and decimals.
- Rates: Frankfurter first, then currency-api, or a manual rate. Refreshed hourly (`cron/fx-refresh.php`). Exchanges pause if the rate is older than 24 h; an automatic rate outside ₱20–200 or more than 10% from the last one is refused.
- Fees 0–5% for buy and sell (default 0.50% for USD, 1.00% for other currencies). The amount is locked to the rate shown.

9 Bills, load and cash-in (SUNIWAY)

Tile	Purpose	SUNIWAY group
UBills	Electricity (AKELCO), water, internet, cable, SSS, Pag-IBIG, PhilHealth…	`bills`
ULoad	Prepaid load for Globe, TM, Smart, TNT, DITO	`eload`
UCash In	Send Credits to GCash, Maya, ShopeePay and other e-wallets	`ecCash`

GET providers
billers, networks, e-wallets
→
POST transactions/preview
provider fee and total shown
→
Debit Credits
then `POST transactions`
→
GET transactions/{id}
until final

- **Successful:** done. **Rejected / Failed:** automatic refund. **Processing:** rechecked; refunded if it later fails.
- **Checking** (no reply in time): Credits stay on hold; Finance resolves it in **Admin › Bills & Load › Needs checking** (Refund or Mark successful).
- Ledger: debit `user_credits`, credit `suniway_payable` (amount + provider fee) and `fee_revenue:<service>` (optional convenience fee).
- The Partner API key is encrypted in settings. "Test connection" counts the billers, products and e-wallets found.

10 Identity: KYC, liveness and Face Login

10.1 KYC

- Available to customers, merchants, drivers and agents: ID front and back, then a guided liveness selfie (look straight, left, right, up, down, blink), then review and send.
- Face detection runs on the phone with MediaPipe. The side to turn to flashes red; correct poses turn green and capture automatically; wrong turns are counted and shown to Admin.
- The "look straight" selfie is compared with the ID photo using face-api. The server re-checks the descriptor distance (default max 0.60) and refuses a mismatch.
- Every submission is reviewed by a person in **Admin › KYC verification**: Approve, Ask to resubmit, or Reject. "Manual" submissions (no automatic check) need extra care.
- Rules (Admin › KYC › Rules): merchants, drivers and agents need approved KYC before approval (on by default); customers only for sending money and UCash In (off by default).

10.2 Face Login (customers)

- Set up in Profile: a short liveness check whose face must match the approved KYC face. The phone gets an HttpOnly, SameSite=Strict device key; only its SHA-256 is stored.
- Login: the server issues random steps and a one-time code valid for 2 minutes, checks step order and timing, and compares the 128-number face descriptor (default max distance 0.45).
- 5 wrong tries lock Face Login on that phone until a password login. Users can turn it off per phone or on all phones.
- Face Login is a convenience login for the person's own phone, not a bank-grade identity proof. The password always works.

> **Privacy**
> Only still photos (KYC) or 128 numbers (Face Login) leave the phone. Photos are re-encoded (no EXIF/GPS), stored with random names in `storage/kyc/`, and shown only to signed-in admins with KYC access. IP address and browser are recorded for audit.

11 Portals

11.1 Admin Panel and roles

Role	Access
Super Admin	Everything, including settings, rates, driver approval and admin users
Finance	Payments and refunds, finance, ledger, settlements, reports, Credits adjustments, driver and agent payouts, currency exchange
Support	Customers, merchant and driver review, tickets, notes, KYC decisions, ride cancellation
Barangay Officer	UBarangay applications for their own barangay only

Every admin action is recorded in the audit log. First-time setup (`/admin/`) creates the Super Admin and then disables itself.

11.2 Merchant Portal and MCTC

- Onboarding: business, owner login, registration numbers, documents, payout account, map pin, KYC → Admin review → Approved.
- After approval: payment QR (optional fixed amount), payments and CSV, payouts, listings (UPass/UGo/ULocal/UEat), orders and bookings, support tickets.
- MCTC Top-up Center (on for every approved merchant by default): scan a customer's or driver's top-up QR, collect cash, enter the receipt and confirm. Cash collected is owed to Ultimate App (`merchants.mctc_due_centavos` , ledger `mctc_merchant_due:ID`) and deducted from the next payout, or recorded as a remittance.

11.3 URide Driver App

- Registration with selfie, license, vehicle (including Boat), plate or boat registration, documents; status Submitted → Under review / Needs info → Approved.
- Online/offline toggle, live request feed with beep, accept/skip, navigation, trip steps, self-drive hand-over and return, fleet units, wallet with QR Ph / MCTC / BCash top-ups and payouts.
- The app must stay open with location allowed to receive requests (browsers pause background GPS).

11.4 Agents Portal

- Approved agents get a referral link (`/register.php?ref=AG…`), code and QR. Customers are tagged once at sign-up, for `agents.earn_days` (default 365).
- Default rates: URide 2% of the fare, UEat 3% (earned after a 3-day hold), UPass/UGo/ULocal 5% (earned when completed). Paid by Ultimate App as an expense.
- Master Agents and Sub-Agents (one level). Masters earn an override on top (default 0.5% URide/UEat, 1% bookings).

11.5 UBarangay

Permits paid with Credits, BCash or QR Ph, reviewed by the Barangay Officer, issued with a verification QR (`/verify.php`). Fees post to `brgy_payable` . Off by default.

12 Security

Control	Implementation
Passwords	`password_hash()` / `password_verify()`; separate logins for customers, drivers, merchants, agents and admins
CSRF	Token per session, checked by `verify_csrf()` on every POST
Rate limits	`rate_limit_enforce(action, actor, limit, window)` backed by the `rate_limits` table (logins, bookings, cancellations, transfers…)
SQL injection	PDO prepared statements; dynamic identifiers come from fixed allow-lists
XSS	All output escaped with `e()`; user text such as cancellation reasons is always escaped
Clickjacking	`X-Frame-Options: DENY`; strict Content-Security-Policy on file viewers and camera pages
Private files	`storage/` and `config/` blocked by `.htaccess`; files served only through permission-checked scripts with sandbox CSP
Secrets	API keys encrypted in `app_settings`; never logged or shown after saving; never commit `config/config.php`, `config/app_secret.php` or database dumps
Webhooks	Signature and timestamp (5 min) verification; idempotent settlement; no raw bodies logged
Audit	Admin audit log; ledger entries are append-only

> **Full package zips contain secrets**
> The full deployment zip includes the live `config/config.php` (database password). Share it only with the server owner, and never upload it to a public place.

13 Deployment

13.1 Server requirements

- PHP 8.1+ with PDO MySQL, cURL, GD (image re-encoding), OpenSSL, mbstring; outbound HTTPS.
- MySQL 8 or MariaDB 10.x with InnoDB (tested on MariaDB 10.11).
- HTTPS (needed for camera, GPS, PWA install and Apple sign-in).
- Writable `storage/` (or the folder in `config/storage.php`) and, once, `config/`.

13.2 Steps (Hostinger)

1. Back up the database and the site files (including `config/` and `storage/`).
2. Import migrations in phpMyAdmin (13.3). Every migration is safe to run again.
3. Upload the update zip (changed files only) or the full zip. Keep the live `config/config.php`, `config/app_secret.php` and `storage/`.
4. Configure in Admin › Settings: PayMongo keys, Google Maps key, URide fares, payment methods, cancellation fees, SUNIWAY.
5. Set the PayMongo webhook to `https://YOUR-DOMAIN/webhook.php` (events `payment.paid` and `checkout_session.payment.paid`).
6. Add the cron job: `php /home/<user>/public_html/cron/fx-refresh.php` every hour.
7. Smoke-test on real phones: booking, payment, driver app, merchant orders.

13.3 Migration order

#	File	Adds
1	`schema.sql`	Base tables (new install only)
2	`paymongo, mctc, fees, conversion, service_requests, notifications, ueat_orders, boracay_cash, boracay_cash_reverse, oauth, rate_limits`	Early platform features (see each module guide)
3	`ALL_UPDATES_2026-09.sql`	News, QR payments, Admin platform, UBarangay, in-app QR Ph, URide driver app, MCTC merchants
4	`agents → agents_team`	Agents Portal, Master/Sub-Agents
5	`suniway`	UBills, ULoad, UCash In
6	`kyc`	Identity verification
7	`bcash_transfers`	BCash Send/Receive
8	`fx → fx_currencies`	Currency exchange
9	`uride_places → uride_rental → uride_selfdrive`	Places, Rent a Ride, Self-drive
10	`face`	Face match and Face Login
11	`marketplace`	UGo, ULocal, UEat merchants
12	`uride_fleet → upass_credits`	E-Bike, Van, fleets; UPass bookings
13	`pay_methods`	Pay Cash switch, BCash payments
14	`cancellation`	Cancellation fees and reasons
15	`uride_boat`	Boat vehicle type and fares
16	`uride_pricing_2026_10`	₱50/2 km + ₱25/km (Motorcycle, E-Trike), ₱10 trip fee, ₱50 top-up
17	`uride_rewards · uride_rental_commission`	Driver trip rewards; 15% Rent a Ride commission

13.4 Configuration files

File / variable	Purpose
`config/config.php · DB_HOST, DB_NAME, DB_USER, DB_PASS`	Database connection (environment variables override)
`config/paymongo.php · PAYMONGO_*`	Mode, keys, webhook secrets, `app_url`
`config/app_secret.php`	Generated encryption key for stored secrets — back it up
`config/storage.php`	Optional private uploads folder outside `public_html`
`APP_URL, GOOGLE_*, FACEBOOK_*, APPLE_*`	Social sign-in

14 Testing

CLI test suites run real database transactions against a **copy** of the database and delete everything they create. Never run them on production or against real provider APIs.

```
DB_HOST=localhost DB_NAME=copy_db DB_USER=... DB_PASS=... php tests/uride_boat_test.php
```

Suite	Covers
`uride_test.php`	Fares, night surcharge, commission, top-ups (QR Ph, MCTC, BCash), dispatch radius, first-accept-wins, refunds, payouts, reconciliation
`uride_rental_test.php` · `uride_selfdrive_test.php`	Rentals, scheduling, licenses, deposits, late fees, damage claims
`uride_fleet_test.php` · `uride_places_test.php`	Fleets and availability; place search
`uride_pricing_test.php`	October 2026 fares, ₱10 trip fee, ₱50 top-up and wallet minimum
`uride_boat_test.php`	Boat ride-only rules, fares, dispatch to boat operators, payment
`cancellation_test.php`	Fee rules, free windows, reasons, ledger for URide, bookings, UEat
`pay_methods_test.php`	Pay Cash switch, BCash payments and refunds
`marketplace_test.php` · `p2m_test.php` · `qr_pay_test.php`	Bookings, UEat orders, merchant payments, P2P
`boracay_cash_test.php` · `bcash_transfer_test.php`	Conversions and transfers
`fx_test.php`	Exchange math, rates, fees, limits (needs `fx_fake_api.php` on port 8097)
`suniway_test.php`	Bills, load, cash-in (needs `suniway_fake_api.php` on port 8098)
`paymongo_test.php`	Webhook signatures and settlement (`PM_FAKE_API=1`)
`kyc_test.php` · `face_test.php`	KYC rules and Face Login
`agents_test.php` · `agents_team_test.php`	Commissions, teams, payouts
`mctc_test.php` · `mctc_merchant_test.php`	Cash top-up approvals
`ubarangay_test.php`	Permits, payments, certificates

```
# fake provider servers for local tests
php -S 127.0.0.1:8098 tests/suniway_fake_api.php &
php -S 127.0.0.1:8097 tests/fx_fake_api.php &
```

> **Status at this edition**
>
> All 24 suites passed between 11 and 15 October 2026 (PHP 8.3, MariaDB), together with browser checks of the admin settings, URide booking (including Boat), and the cancel forms.

15 Operations and known limits

15.1 Routine operations

- **Daily:** process driver, agent and merchant payouts (mark paid with the transfer reference); resolve SUNIWAY "Needs checking"; review KYC and driver applications.
- **Weekly:** check Admin › Finance reconciliation (wallets, escrow, MCTC dues); export reports.
- **Backups:** database, `config/` (especially `app_secret.php`) and the storage folder, together.
- **Rates and prices:** starter fares, rental prices, deposits and listing prices are placeholders — confirm them with the LGU, TODA and partners.

15.2 Known limits

- URide distances use straight line × road factor, not routed distance (this also applies to Boat routes).
- UEat delivery is done by the restaurant; no rider dispatch yet.
- Extending a running rental is not built in.
- Liveness and Face Login run on the phone; every KYC is still reviewed by a person.
- QR Ph and bank/e-wallet payouts are manual (Finance marks them paid); no automated payout provider is connected.
- BCT, Buy BCT and Markets pages are "coming soon".
- Stored-value and e-money rules (BSP Circular 1166) should be reviewed with counsel before BCash is offered as redeemable cash.

15.3 Related guides in the package

URIDE-SETUP.md · URIDE-RENTAL-SETUP.md · URIDE-SELFDRIVE-SETUP.md · URIDE-PLACES-SETUP.md · URIDE-BOAT-SETUP.md · CREDITS-AND-RENTAL-FLEET-SETUP.md · MARKETPLACE-SETUP.md · PAYMENT-METHODS-SETUP.md · CANCELLATION-FEES-SETUP.md · BORACAY-CASH-SETUP.md · BCASH-SECTION-SETUP.md · USD-EXCHANGE-SETUP.md · SUNIWAY-SETUP.md · KYC-SETUP.md · FACE-LIVENESS-LOGIN-SETUP.md · AGENTS-PORTAL-SETUP.md · ADMIN-MERCHANT-SETUP.md · MERCHANT-PORTAL-SETUP.md · MCTC-SETUP.md · PAYMONGO-SETUP.md · UBARANGAY-SETUP.md · PWA-SETUP.md · MENU-COMMUNITY-SETUP.md · README.md